



DELHI PUBLIC SCHOOL NEWTOWN

SESSION: 2025-2026

MONDAY TEST

CLASS: IX
SUBJECT: CHEMISTRY

FULL MARKS: 40
DATE: 30/06/2025

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION A (20 MARKS)

Attempt all questions

Question 1

Chose the correct answers from the options given below:

[8]

i) Composition of nuclei of two atoms X and Y are given below:

	X	Y
Number of protons	8	8
Number of neutrons	8	10

The mass number of X and Y and their relation is:

- (A) 16, 18; Isotopes (B) 16, 16; Isobars
(C) 16, 18; Isotones (D) 18, 18; Isomers

ii) A has 9 protons, 9 electrons, and 10 neutrons.

B has 12 protons, 12 electrons and 12 neutrons.

The formula of the compound formed between A and B is:

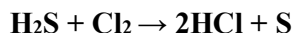
- (A) A_2B (B) BA_2 (C) B_2A_3 (D) AB_4

iii) Assertion (A): Aluminium reacts with both acids and alkalis forming salt and hydrogen.

Reason (R): Aluminium is an amphoteric metal.

- (A) Both A and R are correct, and R is the correct explanation of A
(B) Both A and R are correct, and R is not the correct explanation of A
(C) A is false but R is true
(D) A is true but R is false

iv) In the reaction given below the reducing agent is:

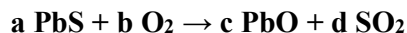


- (A) Chlorine (B) Hydrogen chloride
(C) Hydrogen sulphide (D) Sulphur

v) In the given redox reaction: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

- (A) Zn is reduced to Zn^{2+} (B) Cu^{2+} is oxidised to Cu
(C) Zn is oxidised to Zn^{2+} (D) Cu is oxidised to Cu^{2+}

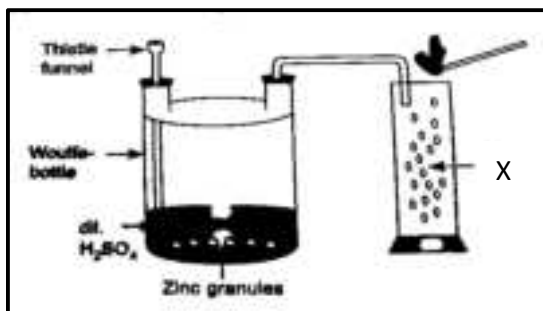
vi) Consider the following chemical equation:



In order to balance this chemical equation, the values of a, b, c and d must be:

- (A) 2, 2, 3, 2 (B) 2, 3, 2, 2 (C) 2, 3, 2, 3 (D) 2, 2, 3, 3

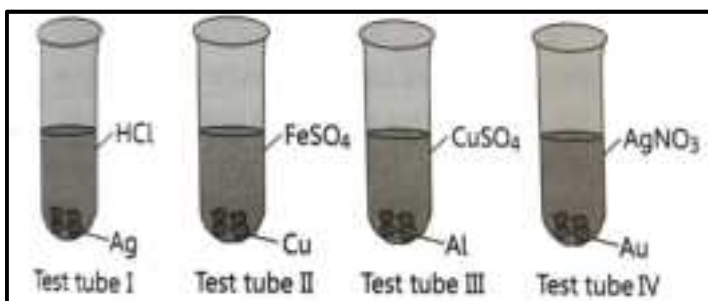
vii) Study the diagram given below:



The method of collecting gas X depicted in the above diagram is incorrect because:

- (A) Gas X is heavier than air
- (B) Gas X is lighter than air
- (C) Gas X forms an explosive mixture with air
- (D) Gas X will extinguish the burning splinter

viii) A student performs the following four experiments:



The correct increasing order of the reactivity of the metals is:

- (A) Au, Ag, Cu, Al
- (B) Au, Al, Ag, Cu
- (C) Al, Ag, Cu, Au
- (D) Cu, Ag, Al, Au

Question: 2

[4]

i) Name the following:

- a) A nitrate which on heating leaves no residue.
- b) A gaseous oxide which is a reducing agent.
- c) A compound which decomposes in presence of sound.
- d) A compound which slows the rate of decomposition of hydrogen peroxide.

ii) Complete the following ionic equations and classify them into oxidation or reduction reactions:

[4]

- a) $\text{Mn}^{7+} \rightarrow \text{Mn}^{5+}$
- b) $\text{Cr}^{3+} \rightarrow \text{Cr}^{6+}$
- c) $\text{Cl}^{-} \rightarrow \text{Cl}$
- d) $\text{O} \rightarrow \text{O}^{2-}$

iii) Write the formula of ammonium nitrate. Calculate the percentage of nitrogen and oxygen in ammonium nitrate. [N = 14, O = 16, H = 1]

[4]

SECTION B (20 MARKS)

Attempt all questions

Question: 3

- i) XCl_2 is the formula of the chloride of metal X.
- State the valency of metal X.
 - Write the formula of the (I) phosphate and (II) zincate of metal X [3]
- ii) Hydrated copper sulphate has the formula $\text{CuSO}_4 \cdot \text{XH}_2\text{O}$ contains 36% water of crystallisation. Determine the value of X. [Cu = 64, S = 32, O = 16, H = 1] [2]

Question: 4

Rahul heated a white crystalline solid P decomposed with a crackling sound, a reddish-brown gas Q with an irritating odour is released along with a colourless odourless gas R which rekindles a glowing splinter and a buff yellow residue S is obtained.

- Identify P and Q.
- What would Rahul observe when gas R is bubbled through alkaline pyrogallol solution.
- Write balanced chemical equation for the reaction when P is heated.
- Rahul observed that residue S dissolved in hot concentrated sodium hydroxide. Which property of residue S did Rahul get to know? [5]

Question: 5

- i) With respect to industrial preparation of hydrogen answer the following:
- Name the catalyst and the promoter used in this process.
 - Write balanced chemical equation for the endothermic reaction.
 - How is unused carbon monoxide removed from hydrogen gas produced?
 - What is water gas? [5]

Question: 6

- i) The electronic configuration of Q^{2-} is 2, 8.
- What is the atomic number of Q?
 - Write the electronic configuration of Q. [2]
- ii) Give reasons for the following:
- The chemical properties of C^{12} and C^{14} are same.
 - Neon is chemically inert.
 - The actual mass of an atom is greater than its mass number. [3]